

Assignment 4

Continuous Random Variables – Part 2

Course: Probability & Statistics

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Total Marks: 50

Due Date: 27 April 2026

Submission Instructions

- Show all steps clearly. Unsupported answers may receive little or no credit.
- Write your solutions neatly and organize them by question number.
- Submit a single PDF file through Canvas.

Question 1 (5 marks)

An absent-minded professor schedules two student appointments for the same time. The appointment durations are independent and exponentially distributed with mean thirty minutes. The first student arrives on time, but the second student arrives five minutes late.

- (a) Express the total time between the arrival of the first student and the departure of the second student in terms of the appointment durations. [2]
- (b) Determine the expected value of this time. [3]

Question 2 (6 marks)

Alvin throws darts at a circular target of radius r and is equally likely to hit any point in the target. Let X be the distance of Alvin's hit from the center of the target.

- (a) Find the probability density function (PDF) of X . [2]
- (b) Compute the mean $\mathbb{E}[X]$. [2]
- (c) Compute the variance $\text{Var}(X)$. [2]

Question 3 (6 marks)

A defective coin minting machine produces coins whose probability of heads is a random variable P with probability density function

$$f_P(p) = \begin{cases} 6p(1-p), & 0 \leq p \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

A coin produced by this machine is selected and tossed repeatedly, with successive tosses assumed independent.

- (a) Find the probability that a coin toss results in heads. [2]
- (b) Given that a coin toss resulted in heads, find the conditional PDF of P . [2]
- (c) Given that the first toss resulted in heads, find the conditional probability of heads on the next toss. [2]

Question 4 (7 marks)

A company makes a certain device. Around 2% of the devices are defective from the start and thus have a lifetime of 0 years. If a device is not defective, its lifetime is exponentially distributed with parameter $\lambda = 2$ (per year). Let X be the lifetime of a randomly chosen device.

- (a) Find the generalized probability density function of X . [2]
- (b) Find $\mathbb{P}(X \geq 1)$. [2]
- (c) Find $\mathbb{P}(X > 2 \mid X \geq 1)$. [1]
- (d) Find $\mathbb{E}[X]$ and $\text{Var}(X)$. [2]

Question 5 (6 marks)

A continuous random variable is said to have a Pareto(x_m, α) distribution with PDF

$$f_X(x) = \begin{cases} \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, & x \geq x_m, \\ 0, & x < x_m, \end{cases}$$

where $x_m, \alpha > 0$.

- (a) Find the cumulative distribution function $F_X(x)$. [2]
- (b) Find $\mathbb{P}(X > 3x_m \mid X > 2x_m)$. [2]
- (c) If $\alpha > 2$, find $\mathbb{E}[X]$ and $\text{Var}(X)$. [2]

Question 6 (6 marks)

The metro train arrives at the station near your home every quarter hour starting at 6:00 a.m. You arrive at the station between 7:10 and 7:30 a.m. Let X be the elapsed time (in minutes) between 7:10 and your arrival, and let Y be the waiting time until you board a train.

- (a) Express the cumulative distribution function (CDF) of Y in terms of the CDF of X . [3]

- (b) Differentiate the result to obtain the PDF of Y . [3]
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Question 7 (7 marks)

Let (X, Y) be the Cartesian coordinates of a randomly chosen point, according to a uniform distribution, in the triangle with vertices at $(0, 1)$, $(0, -1)$, and $(1, 0)$. Find the cumulative distribution function (CDF) and the probability density function (PDF) of $|X - Y|$.

Question 8 (3 marks)

Two points are chosen randomly and independently from the interval $[0, 1]$ according to a uniform distribution. Show that the expected distance between the two points is

$$\mathbb{E}[|X - Y|] = \frac{1}{3}.$$

Question 9 (4 marks)

We start with a stick of length 1. It is broken at a point chosen uniformly, and the piece containing the left end has length X . The process is repeated on this piece, and Y denotes the length of the remaining piece after the second break.

- (a) Find the joint PDF of X and Y . [1]
- (b) Find the marginal PDF of Y . [1]
- (c) Use the PDF of Y to evaluate $\mathbb{E}[Y]$. [1]
- (d) Evaluate $\mathbb{E}[Y]$ by exploiting the relation $Y = X \left(\frac{Y}{X}\right)$. [1]
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